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Safety aware fuzzy longitudinal controller for automated vehicles

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HIGHLIGHTS

- A fuzzy longitudinal controller for vehicles uses fuzzy safety metrics to evaluate the level of safety.
- Data from a real-world experiment with a platoon of 5 vehicles with adaptive cruise control are used for comparison.
- The proposed controller is shown to be safe, comfortable, efficient for traffic flow and string stable in comparison.

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ABSTRACT

Vehicle automation holds significant opportunities in the efforts for safer road traffic. Even lower levels of automation like adaptive cruise control (ACC) systems have great potential. Apart from safety benefits, ACC systems could improve the efficiency of traffic networks. However, contemporary ACC systems are designed to be comfortable, but not guaranteed to be safe. Moreover, according to recent literature studies, they have been found to be conservative and string unstable, with traffic oscillations being amplified when traveling downstream. Due to those issues, severe problems can arise in the traffic flow and the efficiency of road traffic networks may deteriorate. Automating the vehicles' longitudinal movement through ADAS systems offers new opportunities in improving the safety conditions on the roads by designing more consistent controllers. In the present paper, a fuzzy controller is developed for automated longitudinal vehicle control. The controller is validated through simulation and real-world data of ACC systems that are currently available in the market. Using fuzzy surrogate safety metrics, the controller becomes aware of the safety level of any situation and can react properly. Moreover, the driving behavior is efficient for the traffic flow, comfortable and string stable, dampening traffic shockwaves that travel upstream. Results prove the robustness of the proposed solution.

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1. Introduction

According to the “Safe Mobility—a Europe that Protects” fact-sheet, it is estimated that 25,300 people lost their lives on EU roads in 2017 (European Commission, 2018). This is 20% fewer than 2010, but only 2% less than 2016. Moreover, 135,000 people have been seriously injured. The estimated socio-economic cost of road accidents is €120 billion a year. The goal of halving the road fatalities between the years 2010 and 2020 will be difficult to be fulfilled. However, there is strong motivation in creating safer road traffic. The long-term goal is to have close to zero fatalities and serious injuries by 2050 as in the Swedish approach “Vision Zero” (FEVR, 2019). In this effort, vehicle automation is considered to be a significant factor.

Thanks to recent advances in vehicle automation and connectivity, a driver-less, connected and cooperative road transport can be envisioned (Alonso Raposo et al., 2019; Elliott et al., 2019). The best potential for improving transportation networks comes with fully automated vehicles. This corresponds to vehicles with automation Levels 5 or 4 if there are limits on their operational design domain (ODD) (SAE International, 2021). However, achieving automation Levels 5 or 4 with a wide enough ODD may not be possible before 2070 (Shladover, 2016). Currently, only vehicles of automation Levels 1 and 2 are available in the market and only restricted Level 3 technologies are expected to be deployed in the near future.

Fortunately, even lower levels of automation have great potential. It has been shown that using commercially available forward collision warning (FCW) and autonomous emergency braking (AEB) systems can reduce the risk of rear end collisions up to 50% (Cicchino, 2017). Furthermore, it has been shown that the driver acceptance of collision avoidance systems and automation systems of lower levels, such as adaptive cruise control (ACC) can be very high (Eichelberger and McCart, 2016). The case of ACC is of particular interest, as those systems take over the operation of the brakes and throttle, effectively controlling the longitudinal movement of a vehicle. Sensors such as RADAR, cameras or even LIDAR are used to observe the surroundings of the vehicle. The driver usually can choose the desired speed and the desired headway from any vehicle downstream. The controller maintains the desired speed when this is safe. If there is a slower leader vehicle, the speed is decreased, to keep the desired distance, and then increased again when it is safe.

Apart from safety benefits, ACC systems have potential to improve traffic efficiency, forming car-platoons with small inter-vehicle distances and dampening traffic oscillations. Currently, all commercially available vehicles with ACC capabilities are of automation Levels 1 or 2. Hence, the driver is responsible to observe the environment and react in case of a dangerous situation. Therefore, ACC systems are aiming primarily on comfort rather than safety or traffic efficiency. As a result, ACC can be conservative, decreasing the traffic flow, slow to respond and not always capable of reacting to emergency maneuvers from the vehicles downstream.

In the present paper, a fuzzy longitudinal controller is developed. The aim is for the controller to be safe, while being

comfortable and efficient regarding traffic flow. The core of the controller is a fuzzy inference system, composed by five intuitive rules. Three rules take care of achieving and maintaining the desired speed, when this is safe. The fourth rule is responsible for maintaining a gap that would allow an accident to be avoided in the event of a hard deceleration from the leading vehicle. The last rule is responsible for emergency reactions when the vehicle is in critical safety conditions. The antecedents of the last two fuzzy rules are two fuzzy surrogate safety metrics (SSM) that have already been developed, the proactive fuzzy surrogate safety metric (PFS) and the critical fuzzy surrogate safety metric (CFS) (Mattas et al., 2019). Using those fuzzy sets, the controller is aware of the safety condition of the vehicle, so appropriate action can be taken.

Two different configurations of the controller are developed, using different parameter values. Firstly, a comfortable configuration is produced, with timid accelerations and jerks and longer inter-vehicle gaps. Secondly, the controller is tuned to a dynamic configuration with more aggressive accelerations and jerks and smaller inter-vehicle gaps. The bases for comparison are the trajectories of five vehicles platooning in a test truck while using ACC. The same car-platoon is simulated with the proposed controller configurations, to investigate stability, comfort, and traffic flow.

2. Relevant work

2.1. ACC controllers in the literature

The first generation of adaptive cruise control (ACC) has been available in the market for more than two decades (Xiao and Gao, 2010). The commercially available controllers are proprietary, and their details are not available to the public. However, there have been a lot of ACC system designs published in the literature. A detailed review can be found by He et al. (2019). There are three main options for the design of ACC controllers, the proportional-integral-derivative (PID), the model predictive control (MPC) and fuzzy logic.

There are two types of separate controllers, regarding the output. A high-level controller, that is used to recognize the environment of the vehicle and outputs speed or acceleration commands. Then this output goes to a lower level controller that creates the throttle or brake commands as in the work of Shakouri et al. (2011). Controllers with safety guaranties have been developed (Magdici and Althoff, 2017; Nilsson et al., 2016; Plessen et al., 2018; Zhang et al., 2018). To guarantee safety, Magdici and Althoff (2017) calculated the stopping distance of the vehicles involved and used different control strategies for safe and for emergency situations. Furthermore, controllers optimizing fuel consumption or energy demand have been introduced (Akhegaonkar et al., 2016; Chen et al., 2019; Jia et al., 2018; Li et al., 2017; Vajedi and Azad, 2016; Weißmann et al., 2017; Yu and Shi, 2016). These works include controllers for various powertrains, e.g., Vajedi and Azad have worked on plug-in hybrid electric vehicles (Vajedi and Azad, 2016).

In the work of Akhegaonkar et al. (2016), safety constrains are also taken into account. In the work of Yu and Shi (2016),

traffic flow impacts are considered. Moreover, [Ye et al. \(2019\)](#) proposed a controller based on reinforcement learning. Finally, based on the IDM model ([Kesting et al., 2010](#)), a control strategy for ACC has been proposed, which would change the time-gap and acceleration settings of the controller, depending on the traffic conditions, to facilitate traffic flow in motorways ([Manolis et al., 2020](#)).

Fuzzy inference systems (FIS) have also been applied in ACC controllers developed in the literature. [Naranjo et al. \(2003\)](#) developed a Takagi-Sugeno FIS for cruise control and adaptive cruise control. [Tsai et al. \(2010\)](#) developed a Mamdani type ACC, with the speed difference and the headway as inputs. The “or” operation was calculated using the minimum of the antecedents. A different Mamdani controller has been developed by [Merah and Hartani \(2014\)](#), who used the bicycle model to represent the vehicle dynamics. A fuzzy PD plus I controller has been developed for the design of the lower level control by [Prabhakar et al. \(2019\)](#). Another approach has been adapted by [Haroon et al. \(2019\)](#), who used a Mamdani FIS for the gains of a PID controller.

Overall, the indicators used to showcase the advantages of novel ACC controllers have been safety, energy-fuel efficiency and comfort. Traffic flow impacts and string stability are considered less often. A platoon of vehicles car-following is string stable if the magnitude of a perturbation is decreasing when traveling upstream. For example, if the leading vehicle in a car-platoon applies a small deceleration and then recovers its desired speed, every follower vehicle will decelerate less and less. Contrarily, if a car-platoon is string unstable, the perturbation can keep increasing for the following vehicles, creating what is commonly referred to as a stop-and-go wave, or a shockwave. A detailed analysis on string stability and criteria for car-following models can be found in the work of [Ward and Wilson \(2011\)](#).

Of course, safety is vital. However, contemporary commercial ACC controllers are more a comfort feature than a safety feature, and as such they cannot be guaranteed to be collision free. This may have to change, especially when systems of automation Level 3 will be made available. Another point that has to be raised is the complicated relationship between vehicle behavior and traffic flow. Even a small penetration rate of vehicles can have impacts on traffic, as will be shown in the next sub-section. Hence, even fuel-efficient controllers, if not designed in a way to contemplate about the traffic flow, they could deteriorate the traffic network's condition and that would lead to less efficient traffic ([Mamouei et al., 2018](#)). Moreover, string unstable car- platoons can produce uncomfortable and unsafe conditions. Therefore, any proposed ACC system should be investigated for its impact on traffic flow. Finally, until full automation is possible, it is important that the system's function is understandable and its reactions can be predictable by the human driver ([Larsson, 2012](#)). This is not always the case with some machine learning approaches.

A lot of controllers designed in the literature make use of connectivity, to ensure stability and traffic flow. Vehicles can exchange information with the surrounding traffic or the infrastructure using vehicle to vehicle (V2V) or vehicle to infrastructure (V2I) communication. A controller for

connected automated vehicles was shown to improve the traffic flow at merging segments, although even in this case, very large time-gap selection would lead to decreased capacity, lower than that of conventional traffic ([Letter and Elefteriadou, 2017](#)). V2I can also be used to stabilize heterogeneous platoons ([Wang, 2018](#)). Moreover, vehicle coordination is found capable to minimize environmental impacts for platoons in signalized intersections ([Zhao et al., 2018](#)). The proposed model works also for mixed traffic of both human drivers and connected automated vehicles, smoothing the traffic flow. The driving characteristics of surrounding traffic could be estimated using V2V connectivity and used for optimal cooperative control ([Ge and Orosz, 2018](#)). Finally, MPC can also be used for cooperative traffic, with the objective of energy-economy in highways ([Liu et al., 2017](#)). It should be noted that the present paper is presenting a controller's configuration that is beneficial for traffic flow, safe and comfortable, without the use of connectivity capabilities. Connectivity and cooperation have the potential to improve the network conditions even more, but there are no such systems commercially available yet.

2.2. Commercial ACC systems stability and traffic flow

Although vehicles with commercial ACC systems exist in the market, the implementation details are not available to the public as they are proprietary. However, there have been researchers that conducted experiments on commercial vehicles with ACC capabilities. One of the most prominent is the work of [Milanes and Shaldiver \(2014\)](#). They have showed that the specific ACC system used in the vehicles under test showed string instability. Moreover, on another work, [Shladover et al. \(2012\)](#), found that the capacity will probably not be significantly affected by ACC vehicles, as the drivers would select headways similar to those they use when driving conventional vehicles.

On a more recent work, [Makridis et al. \(2019a, b\)](#) estimated the reaction time of a number of ACC systems on different vehicles. The results showed that the observable response time can be slower than expected, closer to the human reaction time. Moreover, the model used to replicate the ACC behavior and its parameter set can directly impact the traffic flow. The reaction time estimations have been repeated for a different set of ACC capable vehicles on an experiment conducted inside a test truck ([Makridis et al., 2020a](#)). In the same work, an investigation on the stability characteristics of an ACC platoon was carried out. The ACC platoon was shown to be unstable, causing large fluctuations in the speed of the vehicles. This can result to inefficient and unsafe traffic.

To investigate the possible impact of vehicles of higher levels of automation on a real network, a simulation experiment has been carried out ([Mattas et al., 2018a, b](#)). The ring road of the city of Antwerp has been modelled using traffic counts data from a weekday morning rush hour. A number of simulations have been carried out, with different market penetration rates of vehicles using an ACC car-following model and different desired time-gap settings. It was shown that they can severely deteriorate the traffic condition of the

network, even for small market penetration rates. On a next step, the fuel consumption was estimated using detailed vehicle models for the vehicle powertrain (Makridis et al., 2020b), showing increased fuel consumption for automated vehicles with no connectivity capabilities.

Summing up, ACC capable vehicles are able to negatively affect the traffic flow, decreasing efficiency and safety, in two major ways. Firstly, very conservative controllers can keep very large distances, decreasing the section's or network's capacity and creating more congestion. Secondly, string unstable car-platoons can amplify perturbations travelling upstream and result to stop-and-go traffic or even accidents. Therefore, it is desirable for a longitudinal controller of an automated vehicle to be safe and string stable in all ranges, without being too conservative.

3. Fuzzy SSMs

Two fuzzy surrogate safety metrics have been developed (Mattas et al., 2019) and are used in the design of the proposed longitudinal controller. The formulation of the fuzzy SSMs is essential for the fuzzy controller. Any car-following situation is characterized as safe or unsafe, or safe with a degree of truth. This is what makes the controller “aware” of the risk level of any situation, so the response is appropriate. Hence, with proper tuning, a controller can be efficient, avoiding over-reactions or being over conservative.

Both fuzzy SSMs have been developed according to the same framework for fuzzy SSMs for rear end collisions (Mattas et al., 2018a, b). According to this framework, for any car-following situation two distances have to be defined, the maximum unsafe distance and the minimum safe distance. Then, the actual inter-vehicle distance is assessed. If it is longer than the minimum safe distance, then the situation is certainly safe. Else, if it is shorter than the maximum unsafe one, the situation is certainly unsafe. Finally, if the actual distance is between the two limits, then the situation is unsafe with a certain degree of truth. Thus, the fuzzy set “unsafe situation” is formulated as follows in Eq. (1).

$$\mu_A(d) = \begin{cases} 1 & 0 \leq d < d_{\text{unsafe}} \\ 0 & d \geq d_{\text{safe}} \\ \frac{d - d_{\text{safe}}}{d_{\text{unsafe}} - d_{\text{safe}}} & d_{\text{unsafe}} \leq d < d_{\text{safe}} \end{cases} \quad (1)$$

where d is the inter-vehicle distance, d_{unsafe} and d_{safe} are the maximum unsafe distance and the minimum safe distance, μ_A is the membership function.

If a situation is not totally safe, a fuzzy interval of the unsafe distance is defined (Mattas et al., 2019). Those intervals can be used as fuzzy SSMs, to ensure a number of attributes such as countable additivity, an attribute necessary for a function to be a measure (Mullakkal-Babu et al., 2017). The fuzzy intervals can be added and averaged, containing also meaningful information about the possible crash severity. Hence, they are important for the comparison of different traffic situations or complete traffic scenarios. In the present work every situation must be assessed individually and the result is used by the controller. Crashes of any severity must

be avoided. Hence, the calculation of fuzzy intervals is not relevant for this work and the fuzzy set “unsafe situation” is to be used directly.

The longitudinal speed controller is responsible to keep safe distance such that “a collision between vehicles can be avoided if the vehicle in front performs an emergency brake”, as defined by the Vienna Convention on Road Traffic (Vanholme et al., 2013). This function is similar to the operation of the ACC in contemporary vehicles. It must be safe and comfortable for the user. Also, it should be efficient as far as this is possible. In other cases, there can be an imminent threat, e.g., a vehicle in front decelerating abruptly. Those cases must also be identified, and proper response must be taken, in order to avoid an accident, even if that includes uncomfortable maneuvers such as hard decelerations. This function is more like the operation of the AEB. Two different fuzzy SSMs are used to identify the two different situations, the proactive fuzzy surrogate safety metric (PFS), and the critical fuzzy surrogate safety metric (CFS). Both have been developed based on the aforementioned fuzzy set “unsafe situation”. They differ in the definition of the maximum unsafe and the minimum safe distance.

3.1. Proactive fuzzy surrogate safety metric

The PFS is used to identify situations in which, even if there is no imminent collision course, in the case that the leading vehicle decelerates hard, the follower vehicle will not be able to avoid the accident. The stopping distance of the leading vehicle is calculated assuming maximum deceleration. The stopping distance of the following vehicle is calculated twice, once for comfortable deceleration and once for maximum. The distance travelled during the response time is also taken into account, assuming constant speed, as the vehicle cannot react in that time interval. If the distance is enough, even by using only comfortable deceleration, the situation is certainly safe. On the other side, if the distance is not enough, even with the hardest deceleration, the distance is certainly unsafe. Hence, the minimum safe and maximum unsafe distances are formulated as follows in Eqs. (2) and (3).

$$d_{\text{safe}}(t) = u_r(t)\tau + \frac{u_r^2(t)}{2b_{\text{rcomf}}} - \frac{u_f^2(t)}{2b_{\text{fmax}}} \quad (2)$$

$$d_{\text{unsafe}}(t) = u_r(t)\tau + \frac{u_r^2(t)}{2b_{\text{rmax}}} - \frac{u_f^2(t)}{2b_{\text{fmax}}} \quad (3)$$

where u_r is the speed of the rear vehicle, u_f is the speed of the front vehicle, t is the time, τ is the reaction time of the rear vehicle, b_{rcomf} is the comfortable deceleration of the rear vehicle, b_{rmax} is the maximum deceleration of the rear vehicle, and b_{fmax} is the maximum deceleration of the leading vehicle. The front vehicle's maximum deceleration must be at least as hard as the maximum deceleration of the rear vehicle.

3.2. Critical fuzzy surrogate safety metric

The CFS metric is responsible for identifying critical safety conditions. The CFS membership value is zero when the speed of the follower vehicle is less or equal to the speed of the

leader vehicle, as there is no collision course. Otherwise, assuming constant speed for the leader vehicle and constant acceleration for the follower during its response time, its stopping distance is calculated. Again, if the accident can be avoided with comfortable deceleration, the situation is certainly safe. If the accident cannot be avoided even with the maximum deceleration of the follower, the situation is considered to be unsafe. Therefore, the minimum safe and maximum unsafe distances are formulated as follows in Eqs. (4)–(9).

$$a'_r(t) = \max(a_r(t), -b_{rcomf}) \quad (4)$$

$$u_r(t + \tau) = u_r a'_r(t) \quad (5)$$

$$\text{If } u_r(t + \tau) \leq u_f(t),$$

$$d_{safe}(t) = d_{unsafe}(t) = \frac{(u_r(t) - u_f(t))^2}{2a'_r(t)} \quad (6)$$

$$\text{else } u_r(t + \tau) > u_f(t),$$

$$d_{new} = \left(\frac{u_r(t) + u_r(t + \tau)}{2} - u_f(t) \right) \tau \quad (7)$$

$$d_{safe}(t) = d_{new} + \frac{(u_r(t) + a'_r(t)\tau - u_f(t))^2}{2b_{rcomf}} \quad (8)$$

$$d_{unsafe}(t) = d_{new} + \frac{(u_r(t) + a'_r(t)\tau - u_f(t))^2}{2b_{rmax}} \quad (9)$$

where a_r is the current acceleration of the rear vehicle and a'_r is the acceleration value limited by the comfortable deceleration. If a vehicle is in a collision course but has already started decelerating, the situation is less unsafe than if it was accelerating. This is taken into account by the constant acceleration assumption for the follower vehicle during its response time.

4. Fuzzy longitudinal control

The longitudinal controller is a Takagi-Sugeno (T-S) inference system (Scherer, 2012). Inputs to the system are the ego-vehicle velocity and acceleration, the leader vehicle velocity, and the inter-vehicle distance. The parameters to be set are the maximum and comfortable deceleration of the ego-vehicle, its reaction time, its desired velocity, and the maximum acceleration of the leader vehicle. The fuzzy inputs are calculated during the fuzzification process. Afterward, the ruleset is evaluated, and the result goes through a T-S aggregation to output the desired acceleration.

4.1. Fuzzification

The fuzzification process maps the crisp values of the inputs to linguistic variables. Linguistic variables such as “small”, “medium” or “very high”, are used to describe a situation and are the antecedents of the fuzzy “IF... Then ...” rules. The present controller is looking at two fuzzy inputs. One is the safety level of the inter-vehicle distance according to the fuzzy

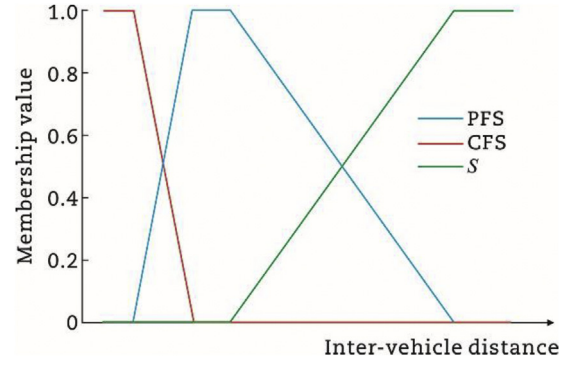


Fig. 1 – Safe, proactively unsafe and critically unsafe linguistic variables.

SSMs. The second is the velocity error, the difference between the actual velocity and the desired velocity.

The inter-vehicle distance can be described to be safe, proactively unsafe or critically unsafe, using the aforementioned fuzzy SSMs. The inter-vehicle distance is determined to be critically unsafe according to the calculation of the CFS safety metric in Eq. (10).

$$CFS(d) = \begin{cases} 1 & 0 \leq d < d_{unsafe} \\ 0 & d \geq d_{safe} \\ \frac{d - d_{safe}}{d_{unsafe} - d_{safe}} & d_{unsafe} \leq d < d_{safe} \end{cases} \quad (10)$$

where $CFS(d)$ is the membership value of the linguistic variable critically unsafe distance.

The inter-vehicle distance is determined to be proactively unsafe according the calculation of the PFS with two additions. There are two safety distance margins d_1 and d_2 equal to 2 m, to keep the minimum distance at stop and to stabilize the controller for low speeds. Therefore, Eqs. (2) and (3) are changed to Eqs. (11) and (12), respectively.

$$d_{safe}(t) = u_r(t)\tau + \frac{u_r^2(t)}{2b_{rcomf}} - \frac{u_f^2(t)}{2b_{fmax}} + d_1 \quad (11)$$

$$d_{unsafe}(t) = u_r(t)\tau + \frac{u_r^2(t)}{2b_{rmax}} - \frac{u_f^2(t)}{2b_{fmax}} \quad (12)$$

$PFS(d)$ is the membership value of the linguistic variable proactively unsafe distance, and the actual $PFS(d)$ is calculated in Eq. (13).

$$PFS_{actual}(d) = \begin{cases} 1 & 0 \leq d < d_{unsafe} \\ 0 & d \geq d_{safe} \\ \frac{d - d_{safe} - d_2}{d_{unsafe} - d_{safe}} & d_{unsafe} \leq d < d_{safe} \end{cases} \quad (13)$$

Moreover, if a distance is critically unsafe, the emergency part of the controller must be activated. Hence, the value of PFS to be used will be the minimum between the calculated PFS and the CFS, effectively using the logical value “the distance is proactively safe and not critically unsafe”, as shown in Eq. (14).

$$PFS(d) = \min(PFS_{actual}(d), 1 - CFS(d)) \quad (14)$$

Finally, the linguistic variable safe inter-vehicle distance (S) is calculated as the negation of the two previous linguistic variables. Thus, the inter-vehicle distance is safe as long as it is neither unsafe proactively nor unsafe critically, as shown in Eq. (15).

$$S(d) = 1 - \text{CFS}(d) - \text{PFS}(d) \quad (15)$$

The shape of the membership function for a representative situation is presented in Fig. 1.

The second fuzzy input concerns the difference of the actual and the desired velocity of the vehicle. When the vehicle is not restricted by safety constraints, it must approach and maintain its desired velocity. The velocity error is the difference between the desired velocity and the actual velocity as shown in Eq. (16).

$$V_e = V_{\text{desired}} - V_{\text{actual}} \quad (16)$$

where V_e is velocity error, V_{desired} is the desired velocity, and V_{actual} is the actual velocity.

Then the linguistic variables describing the velocity error as very negative (N), very positive (P) or almost zero (Z) are designed as presented in Fig. 2. The linguistic variables N and P are trapezoidal while Z is triangular.

4.2. Ruleset

The ruleset of the FIS is composed by a set of “IF ... Then ...” rules where the linguistic variables of the inputs are used as antecedents, producing a particular consequent. For Mamdani type FIS the consequents are also linguistic variables. In T-S type FIS, as the one proposed in the present work, the consequents take specific values (singletons). They could also be linear functions of the inputs.

The proposed controller uses only five intuitive rules and the consequents are specific values of acceleration (or deceleration when it is negative). The firing strength of a rule is the membership value of its antecedents. For the logical operation “and” the t-norm “min” is used. The five rules are:

- (1) If the distance is safe (S) and the velocity error is very negative (N) then decelerate with a' .

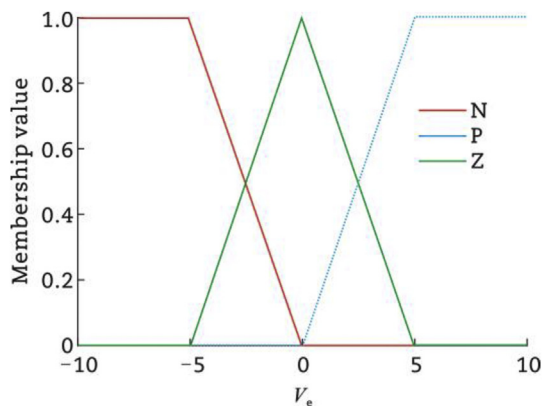


Fig. 2 – Very negative, very positive and almost zero linguistic variables for the velocity error.

- (2) If the distance is safe (S) and the velocity error is very positive (P) then accelerate with b' .
- (3) If the distance is safe (S) and the velocity error is almost zero (Z) then accelerate with 0 (keep constant speed).
- (4) If the distance is proactively unsafe (PFS) then decelerate with c' .
- (5) If the distance is critically unsafe (CFS) then decelerate with d' .

The first three rules are responsible for reaching and maintaining the desired speed when there are no safety constraints. If the speed is higher than desired, the vehicle should decelerate mildly, according to the first rule. If the speed is lower than the desired one, the vehicle should accelerate according to the second rule. If the speed is equal to the desired speed, the vehicle should not accelerate or decelerate, so it will maintain the desired velocity. It must be noted that when for example the velocity is close to the desired speed but a little smaller, both rules 2 and 3 will be activated, with different firing strengths, leading to a mild acceleration.

The fourth and the fifth rules require the vehicle to decelerate if the situation is unsafe. The deceleration forced by each rule is different, as the vehicle is expected to decelerate harder in case of a critically unsafe situation, to avoid an imminent hazard.

The values of the parameters a' , b' , c' , d' will be subject to optimization that will be described in the next chapter.

4.3. T-S aggregation

The firing strength of each of the rules is the membership value of the antecedent proposition. For rules 1–3, the logical operator “and” is combining two linguistic variables and is calculated using the t-norm “min”. Hence, the minimum membership value of the two linguistic variables is the firing strength of the rule. In any case, at least one of the rules must have firing strength greater than zero.

After calculating the firing strength of each rule, denoted f_i for rule i , the acceleration output is calculated according to Eq. (17).

$$\text{acceleration} = \frac{f_1 a' + f_2 b' + f_4 c' + f_5 d'}{f_1 + f_2 + f_3 + f_4 + f_5} \quad (17)$$

where f_i is the firing strength of rule i .

It must be noted that because the antecedents of the rules are fuzzy linguistic variables, taking membership values in the range $[0, 1]$, it is possible that for an instance two or three rules will be activated. Therefore, the acceleration will be a weighted average of the output of each rule, using the firing strength of each rule as a weight. Moreover, the outcome of the third rule is always zero, so it is omitted in the numerator of Eq. (17).

5. Methodology

5.1. Experimental data

To validate the proposed fuzzy controller, its behavior is compared to the behavior of commercial ACC systems. For

this purpose, data from an experiment conducted on a test truck have been utilized (Makridis et al., 2020a). In that experiment, five high-end, ACC capable vehicles were driven in a car-platoon formation around the rural road in the AstaZero test track. All vehicles have been using ACC for longitudinal control during the experiment. In the part of the experiment used in the present work, the leader vehicle has been keeping a constant speed of 70 km/h and when the platoon speed was observed to be stable, it started small perturbations in the form of decelerations and accelerations back to the desired speed. The perturbations were carried out using the ACC system, by changing the desired speed of the leading vehicle, to make sure that the intensity of the deceleration is reproducible and not too hard. Moreover, all the vehicles were using the smaller time-gap option. The data acquisition system used was the RT-Range S multiple target ADAS measurements solution by OxTS (OxTS, 2019) with horizontal speed accuracy around 0.01 m/s and accuracy for the vehicles' position around 0.02 m. The speed profile of the leader vehicle is presented in Fig. 3. The full dataset is openly accessible as part of the OpenACC database of experiments, together with observations from similar experiments conducted in highways (Makridis et al., 2021).

5.2. Model tuning

There are a number of parameters whose values have to be defined. For the calculation of PFS and CFS, the parameters b_{rcomf} , b_{rmax} and b_{fmax} have to be defined. It is noted that the maximum deceleration of the vehicle is not necessarily the maximum deceleration that will be taken into account in the model. More conservative values can be used. Moreover, the values of the parameters a' , b' , c' , d' , which are the outputs of four of the five rules, are to be defined.

The response time τ is assumed to be 0.2 s. While there has been expectation that automated vehicles can run with negligible response delay, experiments have shown that the observable response time of commercial ACC systems is too large to be considered negligible (Makridis et al., 2020a). However, this can be a choice of the manufacturer, trying to provide more comfortable and smoother ride to the passengers, and not a restriction of the hardware or software. Hence, the assumption of 0.2 s to include a small

but not negligible response time is assumed. For different response times the controller has to be tuned again and its properties may change.

To tune the controller and find the optimal values for the seven parameters b_{rcomf} , b_{rmax} , b_{fmax} , a' , b' , c' , and d' , the genetic algorithm (GA) for optimization has been used (Meyer-Baese and Schmid, 2014). The boundaries used for each parameter are presented in Table 1. The parameter d has a physical meaning, as it cannot exceed the hardest possible deceleration for contemporary vehicles. Hence, the parameter d cannot be less than -10 m/s^2 , which is a reasonable limit (Seiniger et al., 2013). The parameter b_{fmax} regarding the maximum possible deceleration of the leader vehicle was chosen conservatively to get values in the interval -13 and -11 m/s^2 .

For the GA, in any generation, a population of individual solutions must be sorted according to their fitness. For the fitness of each solution a number of simulations have been carried out. Firstly, the ACC platoon experiment described in previously was replicated using the proposed controller with each solution of the GA. This is the simulation scenario 1. Moreover, a platoon of 10 vehicles have been simulated, travelling with constant speed of 30 m/s, with the leader vehicle decelerating abruptly. Three runs of the deceleration scenario were carried out with different decelerations of 3, 6 and 12 m/s^2 , denoted as to simulation scenarios 2, 3 and 4, respectively.

For the simulation scenarios 2, 3 and 4 with the abrupt decelerations, it is checked that there is no collision. In case of collision, there is a hard penalty in the fitness of the individual, to avoid converging in such a solution. For scenario 1, again if there is a collision there is a hard penalty for the fitness of the model. Moreover, the controller's behavior has to be assessed. Two different options have been investigated, resulting in two different optimization runs and two different resulting controller configurations, a comfortable and a dynamic one.

Firstly, the controller was tuned for a configuration that can be described as comfortable. The fitness was calculated to minimize the distance between the vehicles in the platoon, the average of the squares of the accelerations and the average of the square of the jerks. The jerk is the first derivative of the acceleration and has been used to measure comfort (Bellem et al., 2018). Therefore, the controller will avoid high positive or negative accelerations and jerks, resulting to a comfortable passenger experience. The fitness of each solution was calculated according to Eq. (18).

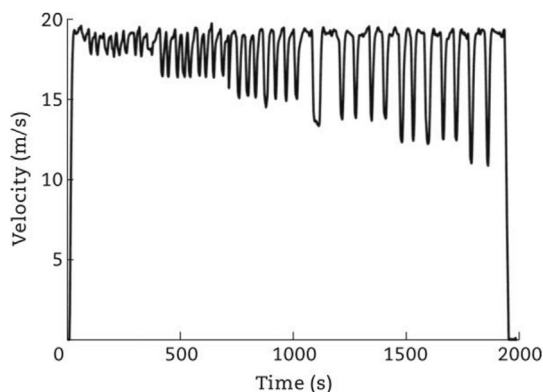


Fig. 3 – The velocity profile of the platoon leader.

Table 1 – Boundaries of parameters.

Parameter	Lower value (m/s^2)	Upper value (m/s^2)
b_{rcomf}	-6	-1
b_{rmax}	-9	-6
b_{fmax}	-13	-11
a'	-6	-2
b'	-4	-1
c'	1	4
d'	-10	-8

$$\text{Fitness} = 1000C + \overline{\text{distance}} + 100 \overline{\text{acceleration}^2} + 1000 \overline{\text{jerk}^2} \quad (18)$$

where C takes the value 0 if there was no crash in the simulation and 1 if there was, $\overline{\text{distance}}$ is the average inter-vehicle distance, $\overline{\text{acceleration}^2}$ and $\overline{\text{jerk}^2}$ are the average of the squares of the acceleration and jerk, respectively.

The second configuration is described as dynamic. The fitness was calculated to minimize only the inter-vehicle distances. In this case, the controller could reach harder accelerations and decelerations, in order to keep small headways and facilitate the traffic flow. The average squared acceleration was also taken into account with a smaller gain. This way, if there were several solutions with similar average inter-vehicle distances, the most comfortable one is preferred. The fitness was calculated according to Eq. (19).

$$\text{Fitness} = 1000C + \overline{\text{distance}} + 10 \overline{\text{acceleration}^2} \quad (19)$$

5.3. Microscopic simulation experiments

For the model tuning and evaluation, a framework for microscopic traffic simulation of the platoon of vehicles has been developed using Python. All the simulations had a 10 Hz frequency, matching the frequency of the data from the experimental campaign. The simulation runs for the same duration as the experimental data. The trajectory of the leader vehicle extracted from the test track experiment was used. Then the behavior of each vehicle in the platoon was simulated to produce the results regarding the fuzzy controller, keeping the initial distance and speed from the experimental data.

The proposed fuzzy controller has been used as a time-delayed car-following model (Sun et al., 2018). The dynamic inputs used are the speed of both vehicles and the inter-vehicle distance. The acceleration was then used, delayed by 0.2 s to account for the controller's response time. The acceleration output of the fuzzy controller was used directly for the simulated vehicle trajectory, without considering possible perturbations because of vehicle dynamics (gear shifts, traction, engine powers, etc.).

5.4. Evaluation method

The tuned models are compared to the real-world observations. The comparison is based on three main dimensions: comfort, stability and traffic flow efficiency. All three dimensions are very important factors and an in-depth investigation is important for each one of those. For each of the dimensions, the fuzzy controller produced will be assessed using proxies, that represent each dimension well and are based on the available data and resources.

Regarding comfort in automated vehicles, it has been realized that the most comfortable driving style for an AV is not necessarily similar to a human driver's behavior (Bellem et al., 2018). According to the same work, a first suggestion would be to keep accelerations and jerks as small as possible. Decelerations that a human driver would use can feel uncomfortable to passengers of an AV. Hence, in the

present work, the acceleration and jerk distributions will be compared between the commercial ACC systems and the proposed fuzzy longitudinal controller, to investigate passenger comfort.

Traffic flow efficiency must be studied under many different conditions. Urban networks, motorways and intersections, signalized or unsignalized can produce different traffic flow characteristics. Moreover, the lateral control logic can significantly impact the traffic condition. Even considering only motorways, a large scale simulation experiment, as in the study of Mattas et al. (2018a, b) can produce meaningful impact. However, the capacity is directly impacted by the desired headway of the vehicles, especially in highway traffic. Hence, smaller headways generally improve the traffic conditions, as higher traffic demand can be served with the same speed. Therefore, the headway of the fuzzy controller will be compared to the headway that ACC systems are keeping.

Concerning string stability, it has to be noted that a vehicle platoon can be stable for some operating conditions and unstable for others. Moreover, string stability can be analyzed as linear, for small perturbations or nonlinear. Nonlinear stability analysis, especially regarding controllers with response time can be challenging. Moreover, comparisons can be made with the commercial ACC systems, only for the conditions of the experiments that have been carried out. Therefore, the fuzzy longitudinal controller will be assessed regarding the stability only for the replicated ACC trajectory, in order to have a meaningful comparison.

6. Results

A fuzzy longitudinal controller has been developed for automated vehicles or vehicles with automated functionalities. The proposed controller is developed to function for the complete range of possible speeds, being safe and efficient regarding to the traffic flow. During the model tuning, two different optimization strategies have been tested, resulting to a comfortable and a dynamic controller configuration. The parameters for both controllers after tuning are presented in Table 2.

6.1. Stability assessment

The results concerning string stability of the fuzzy controller for the given experiment conditions are presented first, as the outcome impacts both comfort and efficiency. Generally, perturbations for the commercial ACC systems have been

Table 2 – Tuned controller parameters.

Parameter	Comfortable (m/s ²)	Dynamic (m/s ²)
$b_{r\text{comf}}$	−5.796	−5.778
$b_{r\text{max}}$	−8.177	−7.020
$b_{f\text{max}}$	−12.576	−11.074
a'	−5.845	−5.383
b'	−2.511	−2.729
c'	1.581	3.637
d'	−8.931	−9.435

shown to be increasing in magnitude for the vehicles upstream (Makridis et al., 2020a). In the case of the fuzzy controller this trend is reversed for both the comfortable and the dynamic setting. A representative part of the trajectory is presented in Fig. 4.

The sections of the experiment presented in Fig. 4, are from the same time interval, following the same leader profile. The platoon leader's speed profile is denoted with the black line and the last vehicle of the platoon the blue one. The speeds of the second, third and fourth vehicles are represented by the more transparent lines for better readability. It is obvious that for the commercial ACC systems, the perturbation is increasing steeply in magnitude, with the last follower's velocity fluctuating between 7.5 and 25.0 m/s. This can have dire consequences regarding both safety and comfort. On the other hand, for both the comfortable and the dynamic configuration of the fuzzy controller, the perturbations are dampened. This could have a positive impact on the traffic networks, even for small market penetration rates, negating the transfer of stop-and-go waves.

6.2. Inter-vehicle distances

Linear or PID based longitudinal controllers usually have an explicit desired time-gap setting. Contrarily, the time-

headway kept by the proposed fuzzy longitudinal controllers is not constant but changes according to the vehicle speeds and the safety requirements. The headway on every instance of the 10 Hz of the simulation is calculated. In Fig. 5(a) the time-gap evolution in time is presented for the first following vehicle. Time-gap is the distance between a vehicle and its leader, divided by the vehicle's velocity. The black dotted line represents the observed commercial ACC systems, whose minimum time-gap setting seems to be close to 1.2 s. There are large oscillations on the time-gap, resulting from the string unstable controller behavior. The comfortable and dynamic fuzzy controllers' headway is represented by the green and blue line, respectively. It is observed that the time-gap is more stable for both cases. Furthermore, both configurations produce smaller average time-gap values than the commercial ACC, while at the same time assuring safety. The comfortable configuration requires larger clearance than the dynamic one, to avoid the need for harder acceleration and deceleration. It is suggested that this will be more appropriate in less congested conditions. When the traffic approaches the point of saturation, the dynamic configuration can be fitting, increasing the capacity and facilitating traffic flow. Moreover, the acceleration term in the dynamic case is much larger than the corresponding comfortable one. This can facilitate the traffic outflow

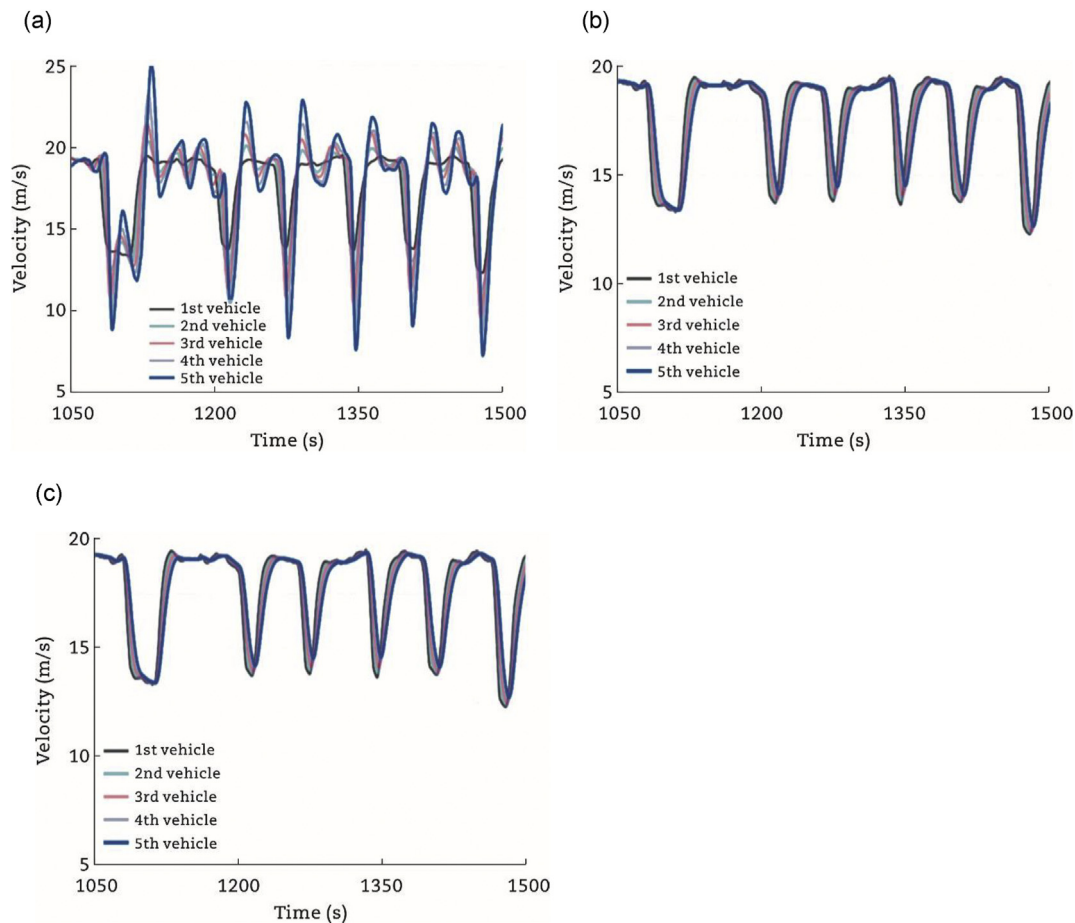


Fig. 4 – Part of the speed trajectory. (a) Commercial ACC. (b) Simulated FIS with comfortable configuration. (c) Simulated FIS with dynamic configuration.

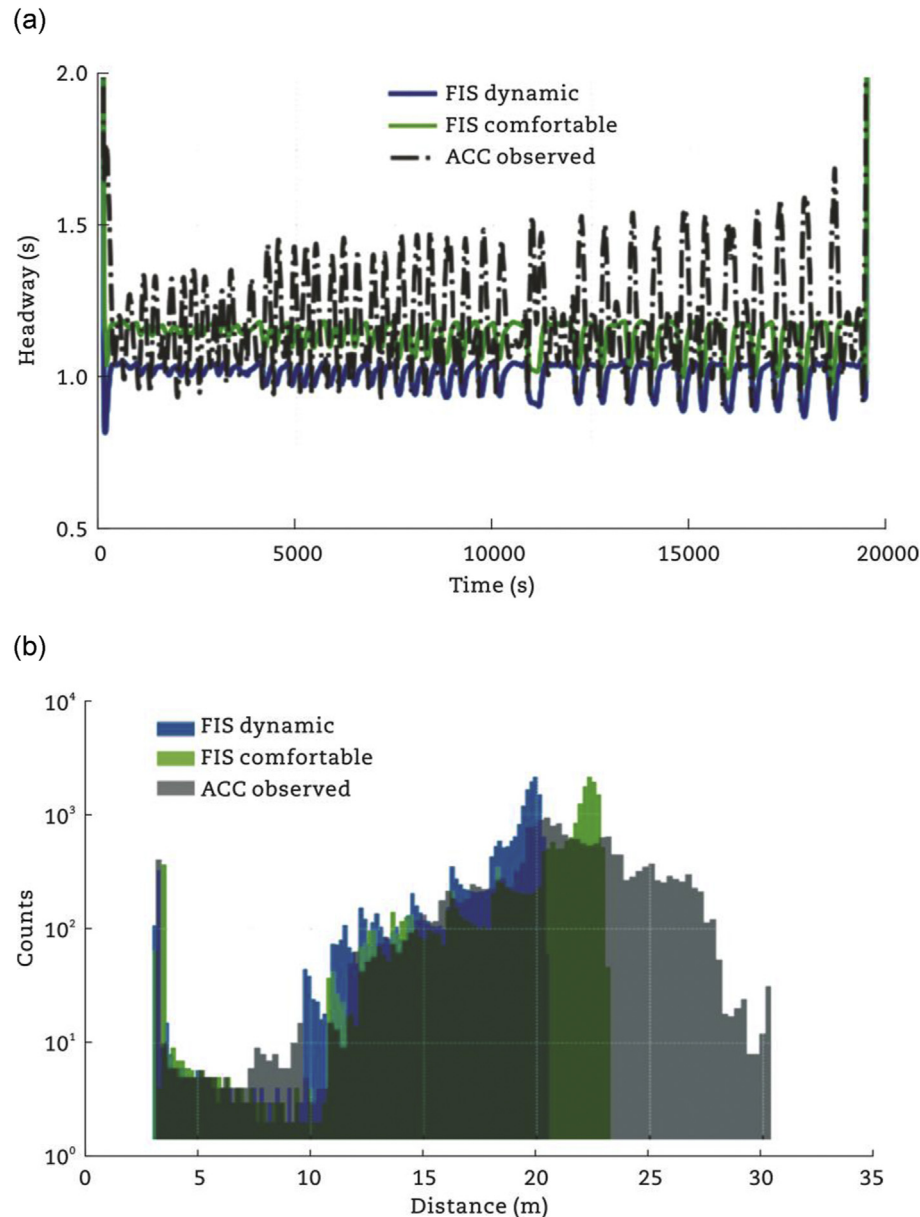


Fig. 5 – Inter-vehicle distances. (a) Evolution of the time-gap between the platoon leader and the first follower. (b) Histogram of the distance between the platoon leader and the first follower.

downstream of a bottleneck and mitigate the capacity drop in such situations (Makridis et al., 2020d). A histogram of the distance headway is presented in Fig. 5(b) in a logarithmic scale for better readability. Here, it is shown that the distance between the leader and the first follower is more than 21 m for a lot of cases of the ACC system, while for the dynamic controller, the distance has been always less. The average distance in the dynamic controller cases has been 17.70, and 19.97 m for the comfortable, comparing to the 20.76 m of the real ACC. Moreover, the maximum value of inter-vehicle distance has been 20.56 m for the dynamic case and 30.39 m on the observed data.

Only the results of the first follower are presented here, as due to instability, the time-gap oscillation is much larger for the vehicles upstream. The second ACC follower follows the

first ACC follower and so on, while the second fuzzy dynamic follower follows the first fuzzy dynamic follower. As seen in Fig. 4, the ACC followers reach very high and low velocities, creating longer distances. Thus, comparing the trajectories of those followers would not be a fair comparison and would be biased against the ACC controller.

6.3. Acceleration and jerk

The relative comfort for the controllers is assessed by the distribution of the accelerations and jerks. Again, the distributions of only the first follower are presented, as due to the unstable behavior of the ACC platoons, the vehicles upstream have much larger speed fluctuation and thus are less comfortable.

In Fig. 6(a) the acceleration distribution is presented for the ACC system in gray, the dynamic fuzzy controller in blue and the comfortable in green. Again, the scale is logarithmic. In Fig. 6(b) using the same notation, the jerk distribution is presented. From Fig. 6 it is obvious that the fuzzy longitudinal controller has been using milder accelerations and jerks, thus producing a smoother ride and creating a more comfortable passenger experience. Additionally, both the acceleration and the jerk values of the proposed controller are within the boundaries of the observed. This does not imply that the trajectory can be exactly replicated by a given vehicle necessarily. The powertrain and vehicle dynamics would

have to be investigated in detail for any given vehicle. However, the fact that the accelerations and jerks are within the limits, and more timid than the observed, shows that the controller is promising for real applications.

The standard deviation of the acceleration for both the dynamic and comfortable simulations is 0.29, and 0.39 m/s^2 in the observed data, signifying a higher variance. Furthermore, the 1st percentile has been -0.94 and -0.95 m/s^2 for the dynamic and comfortable configurations and -1.29 m/s^2 in the observed data.

A comparison between the comfortable and dynamic configurations shows that the distribution of acceleration is

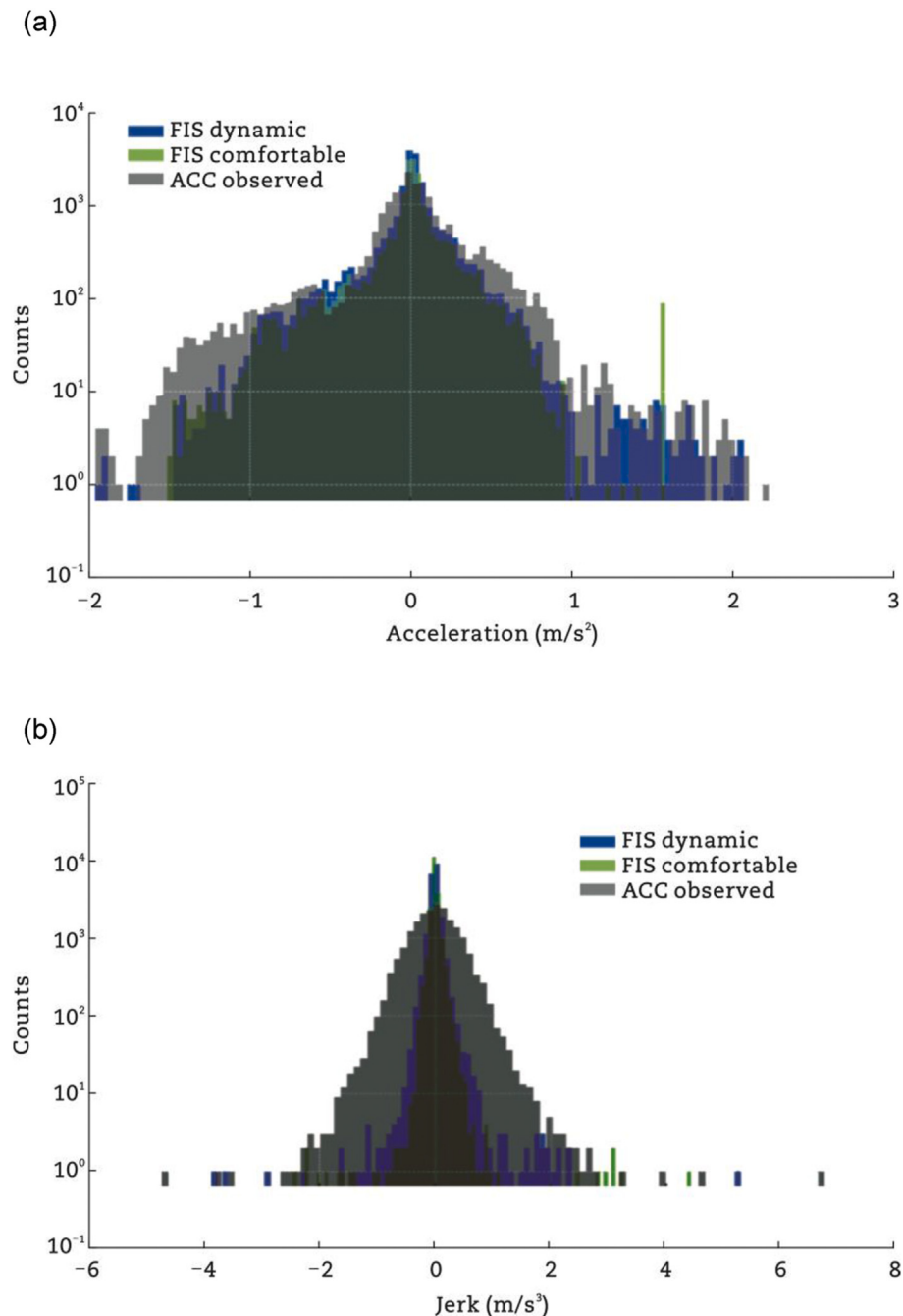


Fig. 6 – Acceleration and jerk. (a) Acceleration distribution of the first follower. (b) Jerk distribution of the first follower.

similarly around zero. The obvious difference comes with accelerations between 1 and 2 m/s², which are very rare for the comfortable configuration, while not as uncommon for the dynamic. In Fig. 6(b) the difference is more obvious with the comfortable configuration having a jerk distribution that is narrower around zero and always within the absolute 5.00 m/s³ limit. The standard deviation of the jerk value has been 0.14 and 0.11 m/s³ for the dynamic and comfortable cases and 0.42 m/s³ in the observed data. Hence, the comfortable configuration is shown to result to a smoother trip, while the contemporary ACC vehicles in the test-track experiments resulted in bumper rides.

7. Conclusions and future work

In the present paper a fuzzy longitudinal vehicle controller is proposed. Two fuzzy safety metrics that have already been developed are used for the controller to be aware of unsafe situations, may they be emergency situations or proactively unsafe, so the aggressiveness of the response can be appropriate.

Data from a real-world experiment with a platoon of 5 ACC vehicles are used. The experiment is replicated in simulation for two different configurations of the fuzzy controller, a comfortable and a dynamic one. The simulated trajectories are compared to the observed ones regarding string stability, comfort and impacts on the traffic flow.

For the string stability, we observe the magnitude of the perturbations on the speed, as they are travelling downstream. It is shown that for the ACC vehicles of the experiment, the magnitude of the perturbations is increasing. This string unstable behavior can create problems regarding comfort, safety efficiency and traffic flow, with the propagation of stop-and-go waves. On the contrary, both configurations of the proposed fuzzy controller are shown to be string stable for the same leader trajectory, dampening any perturbation created by the leader vehicle.

To assess comfort, the distribution of the accelerations and jerks is presented for the first follower vehicle. The simulated vehicles abiding by the fuzzy controller have used smaller in absolute value accelerations and jerks, both for the comfortable and dynamic configuration. This also shows that the vehicle dynamics necessary for the proposed controller are realistic.

Regarding the impact on the traffic flow, it is shown that the inter-vehicle gaps created by the proposed controller are smaller. Hence, the road capacity can be increased. Adding to that the string stable behavior, it can be concluded that the proposed fuzzy controller can be beneficial for the condition of the road networks.

Conflict of interest

The authors do not have any conflict of interest with other entities or researchers.

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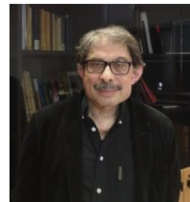
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